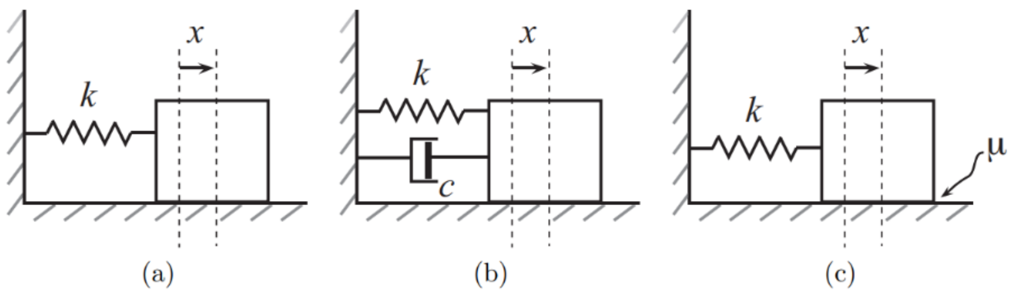
Problem 1

Problem 1



Plot the 2D phase portrait for the three spring-mass systems shown, identifying all important features; the generalized coordinate, x, is the stretch in the spring.

(a) Sliding without friction along the floor.

(b) With a linear damper $F = -b\dot{x}$.

(c) With Coulomb (kinetic) friction due to contact between mass and floor. Coulomb friction can be found by $F = -\mu N \frac{\dot{x}}{|\dot{x}|}$.

(a) Sliding without friction

With x the stretch in the spring, $V = \frac{1}{2}kx^2$ (the natural-length offset $-\frac{1}{2}kx_0^2$ is just a constant and drops out of the dynamics):

$$L = \frac{1}{2}M\dot{x}^2 - \frac{1}{2}kx^2$$

Euler–Lagrange gives the equation of motion:

$$M\ddot{x} + kx = 0$$

State space, with $v = \dot{x}$:

$$\dot{x} = v, \qquad \dot{v} = -\frac{k}{M}x$$

Equilibrium: $(x, v) = (0, 0)$.

$$J = \begin{bmatrix} 0 & 1 \\ -\dfrac{k}{M} & 0 \end{bmatrix}, \qquad \tau = \operatorname{tr} J = 0, \qquad \Delta = \det J = \frac{k}{M}$$

$$\lambda = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2} = \pm\sqrt{\frac{k}{M}}\, i$$

The eigenvalues are purely imaginary, so the origin is a **center**. Trajectories are closed orbits (ellipses in the x–v plane) that encircle the origin: with no dissipation, the mass oscillates forever at constant energy.

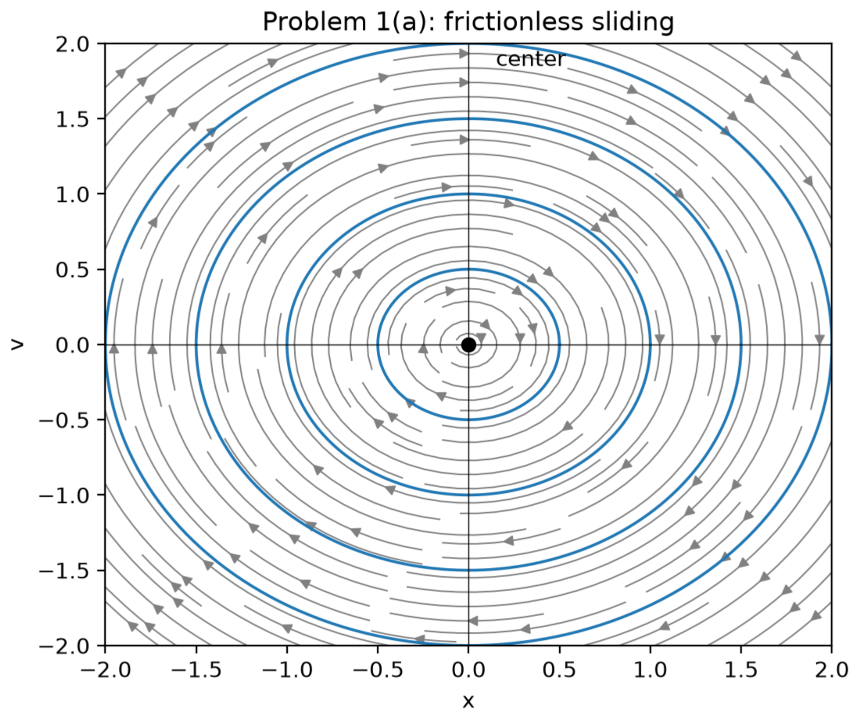


Figure 1: Phase portrait for Problem 1(a): frictionless sliding.

(b) With a linear damper

Newton's second law:

$$-kx - b\dot{x} = M\ddot{x} \qquad \Longrightarrow \qquad \ddot{x} = -\frac{b}{M}\dot{x} - \frac{k}{M}x$$

State space:

$$\dot{x} = v, \qquad \dot{v} = -\frac{b}{M}v - \frac{k}{M}x$$

Equilibrium: $(x, v) = (0, 0)$.

$$J = \begin{bmatrix} 0 & 1 \\ -\frac{k}{M} & -\frac{b}{M} \end{bmatrix}, \qquad \tau = -\frac{b}{M}, \qquad \Delta = \frac{k}{M}$$

Since $M, b, k > 0$, $\tau < 0$ always $\Rightarrow$ the equilibrium is always stable. The character of that stable equilibrium depends on the sign of the discriminant $\tau^2 - 4\Delta$:

$$\frac{b^2}{M^2} - \frac{4k}{M} = 0 \qquad \Longrightarrow \qquad b_{cr} = 2\sqrt{kM}$$

$b > 2\sqrt{kM}$: overdamped, **stable node**

$b = 2\sqrt{kM}$: critically damped, **degenerate (improper) node** — the boundary case

$b < 2\sqrt{kM}$: underdamped, **stable spiral**

At $b = b_{cr} = 2\sqrt{kM}$ the discriminant $\tau^2 - 4\Delta$ vanishes and the two eigenvalues coalesce into the repeated root $\lambda = -b_{cr}/(2M) = -\sqrt{k/M}$: the two straight eigendirections of the overdamped node merge into one, and trajectories approach the origin tangent to that single direction without oscillating — the fastest possible non-oscillatory decay, sitting exactly on the boundary between the node and the spiral.

Eigenvector directions, from

$$\begin{bmatrix} -\lambda & 1 \\ -\dfrac{k}{M} & -\dfrac{b}{M} - \lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \qquad \Longrightarrow \qquad v_1 \lambda = v_2,$$

i.e. $v = \lambda x$ along each eigendirection, with $\lambda < 0$ for both roots in the overdamped case.

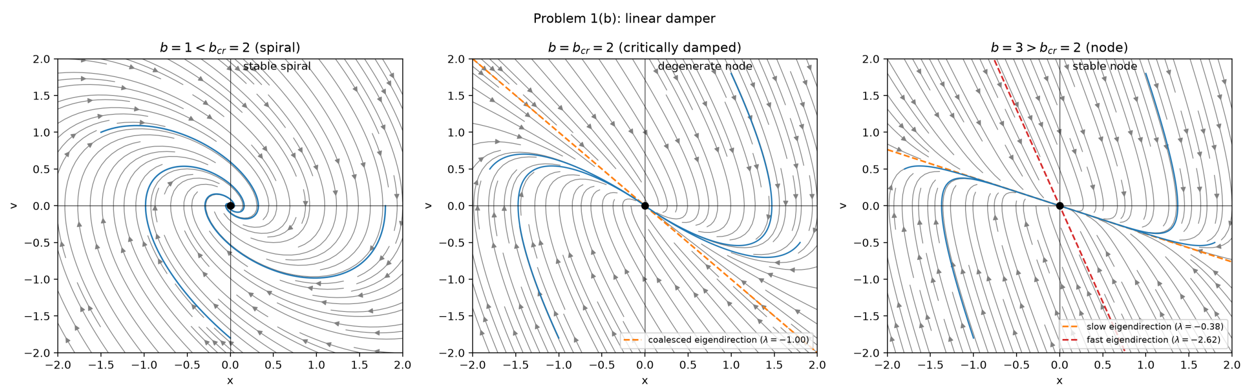


Figure 2: Phase portraits for Problem 1(b): underdamped ($b < 2\sqrt{kM}$, stable spiral, left), critically damped ($b = 2\sqrt{kM}$, degenerate node with the coalesced eigendirection, center), and overdamped ($b > 2\sqrt{kM}$, stable node with fast/slow eigendirections, right).

(c) With Coulomb friction

For $\dot{x} > 0$:

$$M\ddot{x} = -kx - \mu N = -k\left(x + \frac{\mu N}{k}\right) \qquad \Longrightarrow \qquad \text{zero at } x = -\frac{\mu N}{k}$$

For $\dot{x} < 0$:

$$M\ddot{x} = -kx + \mu N = -k\left(x - \frac{\mu N}{k}\right) \qquad \Longrightarrow \qquad \text{zero at } x = \frac{\mu N}{k}$$

The system behaves like an undamped spring-mass oscillating about a fictitious center at $\mp\mu N/k$, depending on the sign of the velocity. In the phase plane the trajectory is a sequence of nested half-circles, alternating about the two centers $\pm\mu N/k$, that shrink by a constant amount each swing until the amplitude falls inside the "dead band" $|x| \leq \mu N/k$ — a whole line of fixed points, not an isolated one, since static friction can hold the mass anywhere in that range — where the mass sticks and motion stops in finite time.

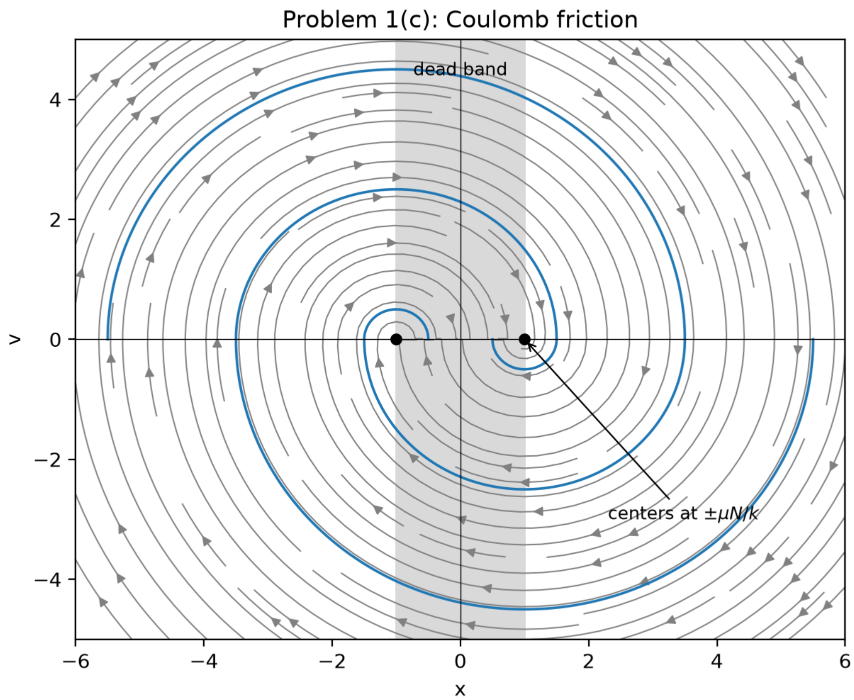


Figure 3: Phase portrait for Problem 1(c): Coulomb friction.

Problem 2

Problem 2

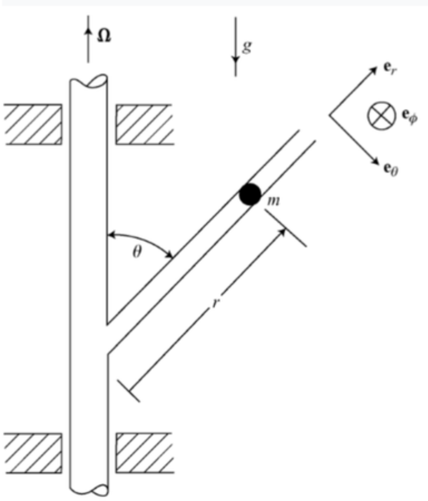
Reconsider the rotating bead of mass m from Homework 1. It is constrained to slide without friction in the tube which is welded to a vertical post at the fixed angle θ. The post rotates with constant angular velocity Ω.

(a) Find a differential equation for the variable r when $\theta = 60^\circ$, using Lagrange's equations this time.

(b) Non-dimensionalize the equation of motion.

(c) Plot the phase portrait of the non-dimensionalized system, identifying all important features. Only show the physically relevant portion of the phase space.

(d) If the bead starts at $r_0 = \frac{4g}{3\Omega^2}$, what minimum initial inward speed $\dot{r}_0$ does it need in order to eventually reach the bottom of the tube, r = 0?
Note: You don't need to directly solve for $r(t)$ to do this. Instead, consider the Hamiltonian (the more general definition, not the total energy), which is a constant of motion, and find the initial conditions that have the same value of H as the equilibrium. This will find the location of the stable manifold at $r = \frac{4g}{3\Omega^2}$.



(a) Equation of motion via Lagrange's equations

Generalized coordinate: x (the bead's position along the tube, i.e. r). The post angle θ is fixed by the geometry (welded post), so we write it as the constant θ_0 throughout, with $\theta_0 = 60°$ only plugged in numerically at the end.

$$\vec{r} = x\,\hat{e}_r, \qquad \dot{\vec{r}} = \dot{x}\,\hat{e}_r + x\,\dot{\hat{e}}_r$$

The post spins about the fixed vertical axis $\hat{\imath}_3$ at rate $\dot{\phi} = \Omega$, carrying $\hat{e}_r$ with it, so $\dot{\hat{e}}_r = \Omega\,\hat{\imath}_3 \times \hat{e}_r = \Omega \sin\theta_0\,\hat{e}_\phi$ (the standard result for a unit vector making the fixed angle θ_0 with the spin axis $\hat{\imath}_3$), where $\hat{e}_\phi$ is the azimuthal direction shown in the figure. Since $\hat{e}_r \perp \hat{e}_\phi$:

$$\dot{\vec{r}} = \dot{x}\,\hat{e}_r + \Omega x \sin\theta_0\,\hat{e}_\phi \qquad \Longrightarrow \qquad |\dot{\vec{r}}|^2 = \dot{x}^2 + \Omega^2 x^2 \sin^2\theta_0$$

$$T = \frac{1}{2}M\left(\dot{x}^2 + \Omega^2 x^2 \sin^2\theta_0\right), \qquad V = Mgx\cos\theta_0$$

$$L = T - V = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}M\Omega^2 x^2 \sin^2\theta_0 - Mgx\cos\theta_0$$

Euler–Lagrange: $\dfrac{d}{dt}\left(\dfrac{\partial L}{\partial \dot{x}}\right) - \dfrac{\partial L}{\partial x} = 0$

$$M\ddot{x} - \Omega^2 M \sin^2\theta_0\, x + Mg\cos\theta_0 = 0$$

(b) Non-dimensionalization

$$\ddot{x} = \Omega^2 \sin^2\theta_0\, x - g\cos\theta_0$$

Let $x = Lu$, $t = T\varsigma$, so $\dfrac{d^2x}{dt^2} = \dfrac{L}{T^2}\dfrac{d^2u}{d\varsigma^2}$:

$$\frac{L}{T^2}u'' = \Omega^2\sin^2\theta_0\, Lu - g\cos\theta_0 \qquad \Longrightarrow \qquad u'' = T^2\Omega^2\sin^2\theta_0\, u - \frac{T^2 g\cos\theta_0}{L}$$

Choose the scalings so both coefficients become 1:

$$T^2\Omega^2\sin^2\theta_0 = 1 \;\Rightarrow\; T = \frac{1}{\Omega\sin\theta_0}, \qquad\qquad \frac{T^2 g\cos\theta_0}{L} = 1 \;\Rightarrow\; L = \frac{g\cos\theta_0}{\Omega^2\sin^2\theta_0}$$

$$\ddot{u} = u - 1$$

This nondimensional equation holds for any θ_0; only the scalings T and L depend on it. At $\theta_0 = 60^\circ$ ($\sin^2\theta_0 = 3/4$, $\cos\theta_0 = 1/2$):

$$L\Big|_{\theta_0=60^\circ} = \frac{\frac{1}{2}g}{\Omega^2\cdot\frac{3}{4}} = \frac{2}{3}\frac{g}{\Omega^2}$$

(c) Phase portrait of the non-dimensionalized system

Let $v = u'$:

$$u' = v, \qquad v' = u - 1$$

Equilibrium: $(u, v) = (1, 0)$.

$$J = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \tau = 0, \qquad \Delta = -1 \qquad \Longrightarrow \qquad \textbf{saddle point}$$

$$\lambda = \pm 1$$

Eigendirections through the saddle: $v = u - 1$ (unstable, $\lambda = +1$) and $v = -(u-1)$ (stable, $\lambda = -1$).

Conserved quantity: since $v\,dv = (u-1)\,du$,

$$\frac{1}{2}v^2 - \frac{1}{2}(u-1)^2 = h = \text{const}$$

Since $u = r/L \geq 0$ physically, only the half-plane $u \geq 0$ of this phase portrait is physically relevant.

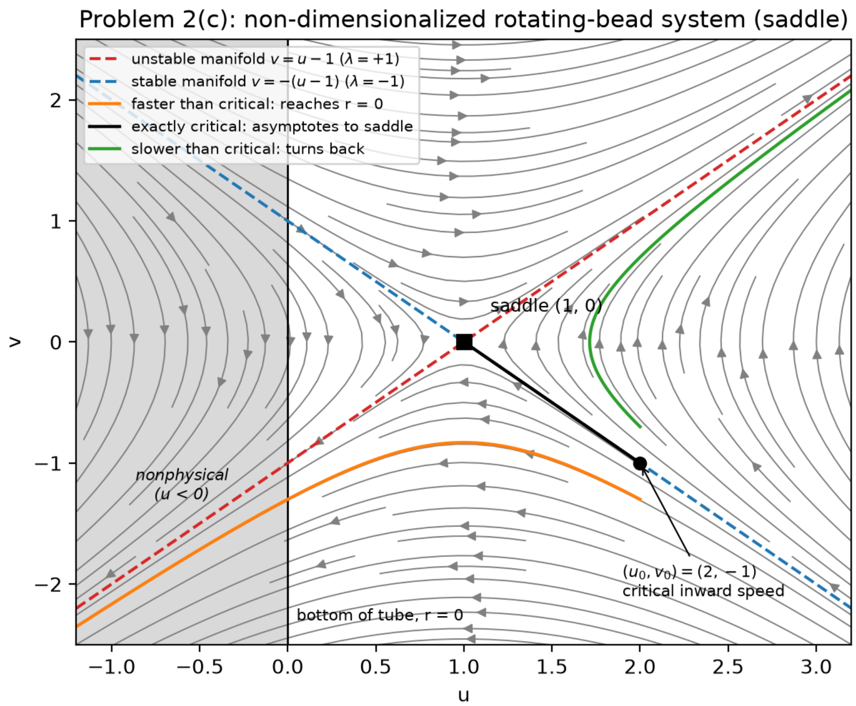


Figure 4: Phase portrait of the non-dimensionalized rotating-bead system: saddle at $(1, 0)$, physically relevant region $u \geq 0$ only.

(d) Minimum initial inward speed

Given $r_0 = \dfrac{4g}{3\Omega^2} = Lu_0$ and $L = \dfrac{2}{3}\dfrac{g}{\Omega^2}$:

$$u_0 = 2$$

The Hamiltonian. The problem asks specifically for the Hamiltonian (the Legendre transform, not simply $T + V$), since the constraint driving the tube is rheonomic. With one generalized coordinate x:

$$H = \dot{x}\,\frac{\partial L}{\partial \dot{x}} - L = \dot{x}(M\dot{x}) - \left(\frac{1}{2}M\dot{x}^2 + \frac{1}{2}M\Omega^2 x^2 \sin^2\theta_0 - Mgx\cos\theta_0\right)$$

$$H = \underbrace{\frac{1}{2}M\dot{x}^2}_{T_2} - \underbrace{\frac{1}{2}M\Omega^2 x^2 \sin^2\theta_0}_{T_0} + \underbrace{Mgx\cos\theta_0}_{V} \neq E = T + V,$$

since the velocity-independent piece of the kinetic energy, T_0, enters H with a minus sign instead of a plus. H is conserved because $\partial L/\partial t = 0$ (Ω, θ_0 are constants, so L has no explicit time dependence). The constraint here is rheonomic, but conservation survives it for this system specifically: the drive is axisymmetric, entering L only through the constant Ω and never through t directly.

Nondimensionalizing with $x = Lu$, $\dot{x} = (L/T)v$ and the scalings $\Omega^2 \sin^2\theta_0 = 1/T^2$, $Mg\cos\theta_0 = ML/T^2$ from (b):

$$H = \frac{ML^2}{T^2}\left(\frac{1}{2}v^2 - \frac{1}{2}u^2 + u\right) = \frac{ML^2}{T^2}\left[\underbrace{\frac{1}{2}v^2 - \frac{1}{2}(u-1)^2}_{h} + \frac{1}{2}\right]$$

So, up to the positive overall scale ML^2/T^2 and an additive constant $\frac{1}{2}$, H reduces to exactly the conserved quantity $h = \frac{1}{2}v^2 - \frac{1}{2}(u-1)^2$ found in (c) by integrating $v\,dv = (u-1)\,du$ — confirming that quantity is (proportional to, up to offset) the Hamiltonian. At the saddle $(1,0)$, $h=0$; this is also the value of h everywhere on the stable/unstable manifolds $v = \mp(u-1)$.

Finding the threshold. At $u_0 = 2$: $h = \frac{1}{2}v_0^2 - \frac{1}{2}$. The level curves of h are hyperbolas; only the branch with $h > 0$ actually connects the far side $u_0 = 2$ down through the saddle region to $u = 0$ with nonzero speed the whole way. At exactly $h = 0$ the bead is riding the stable manifold itself: it asymptotes to $(u,v) \to (1,0)$ as $\varsigma \to \infty$ and never actually reaches $u = 1$, let alone $u = 0$. So the requirement is the strict inequality

$$h > 0 \qquad \Longrightarrow \qquad v_0^2 - 1 > 0 \qquad \Longrightarrow \qquad v_0 < -1 \ \text{ (inward branch).}$$

Consequently $v_0 = -1$ is not itself an attainable initial speed — it is the infimum of the inward speeds that work, approached but never reached, since at $|v_0| = 1$ the bead only creeps toward $r = L$ and never gets to the bottom of the tube.

$$\frac{L}{T} = \frac{g\cos\theta_0/(\Omega^2\sin^2\theta_0)}{1/(\Omega\sin\theta_0)} = \frac{g}{\Omega\tan\theta_0} \qquad \Longrightarrow \qquad \dot{x}_0 = \frac{L}{T}v_0 = -\frac{g}{\Omega\tan\theta_0}$$

The greatest lower bound on the inward speed (an infimum, not an attained minimum) is

$$\dot{x}_0 = -\frac{g}{\Omega\tan\theta_0} = -\frac{g}{\sqrt{3}\,\Omega} \qquad (\theta_0 = 60^\circ)$$

Problem 3

Problem 3

Consider a mass-spring system, with $m = 1$ and $k = 1$, that is experiencing a damping force of $F = \mu(1 - x^2)\dot{x}$.

(a) Write its equation of motion, $\ddot{x} = f(x, \dot{x}; \mu)$.

(b) Convert to state-space by using $v = \dot{x}$ to write the two first-order ODEs $\dot{x} = f_1(x, v; \mu) = v, \dot{v} = f_2(x, v; \mu)$.

(c) Choose $\mu = -1$ and draw the phase portrait. Where is the one equilibrium in the system?

(d) Linearize the system about $(x, v) = (0, 0)$ using the Jacobean,

$$\begin{bmatrix} \dot{x} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial v} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial v} \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix}$$

.

(e) Find the eigenvalues of the Jacobean. Is the equilibrium point stable for $\mu = -1$?

(f) Now choose $\mu = 1$ and find the eigenvalues of the Jacobean matrix again. Is the equilibrium point stable now?

(g) Draw the phase portrait again. What has changed about the system?

(a) Equation of motion

With $m = 1$, $k = 1$:

$$\ddot{x} = -x + \mu(1 - x^2)\dot{x}$$

(b) State space

$$\dot{x} = v, \qquad \dot{v} = -x + \mu(1 - x^2)v$$

(c) $\mu = -1$

$$\dot{x} = v, \qquad \dot{v} = -x - v(1 - x^2)$$

Fixed point: $(0, 0)$.

Under time reversal $t \to -t$ (so $\dot{x} \to -\dot{x}$ while $x, \ddot{x}$ are unchanged), the equation $\ddot{x} = -x + \mu(1 - x^2)\dot{x}$ maps into itself with $\mu \to -\mu$: the $\mu = -1$ system is the exact time-reversal of the $\mu = 1$ system studied in (f)–(g). Part (g) finds that $\mu = 1$ is the standard Van der Pol oscillator, with an unstable origin (confirmed by eigenvalues in (f)) surrounded by a stable limit cycle of amplitude ≈ 2. Time-reversing swaps the stability of *both* features at once, so for $\mu = -1$ the origin is a locally stable spiral (confirmed by the eigenvalues in (e)) surrounded by an **unstable** limit cycle of the same amplitude, ≈ 2.01 numerically. The origin's basin of attraction is only the interior of this unstable cycle — initial conditions outside it escape to infinity in finite time rather than spiraling in, which is why the figure below only seeds trajectories inside the cycle and draws the cycle itself by integrating backward in time from near the origin.

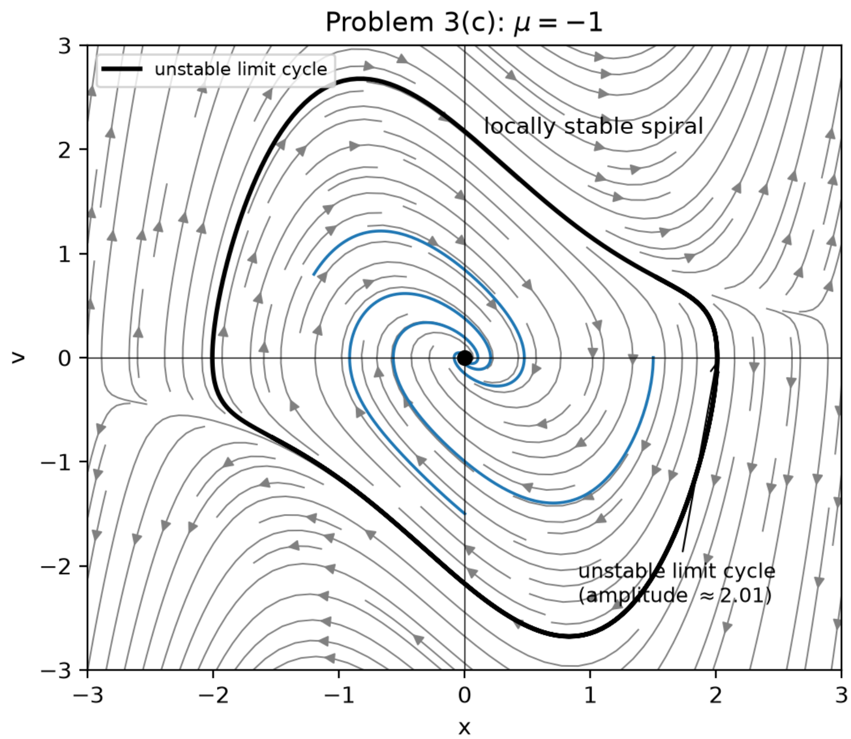


Figure 5: Phase portrait for Problem 3(c): $\mu = -1$. The origin is only *locally* stable: its basin is bounded by an unstable limit cycle at amplitude ≈ 2.01.

(d) Jacobian

With $f_1 = v$ and $f_2 = -x + \mu(1 - x^2)v$:

$$\frac{\partial f_2}{\partial x} = -1 - 2\mu xv, \qquad \frac{\partial f_2}{\partial v} = \mu(1 - x^2)$$

$$J = \begin{bmatrix} 0 & 1 \\ -1 - 2\mu xv & \mu - \mu x^2 \end{bmatrix}$$

At $\mu = -1$: $J = \begin{bmatrix} 0 & 1 \\ -1 + 2xv & -1 + x^2 \end{bmatrix}$, and at the equilibrium $(0, 0)$:

$$J\Big|_{(0,0)} = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}, \qquad \tau = -1, \qquad \Delta = 1$$

(e) Eigenvalues for $\mu = -1$

$$\lambda = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2} = \frac{-1 \pm \sqrt{1 - 4}}{2} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$$

Re$(\lambda) < 0$ for both roots $\Rightarrow$ **stable spiral**. The equilibrium is stable for $\mu = -1$.

(f) Eigenvalues for $\mu = 1$

$$J\Big|_{(0,0)} = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}, \qquad \tau = 1, \qquad \Delta = 1$$

$$\lambda = \frac{1}{2} \pm \frac{\sqrt{3}}{2} i$$

$\mathrm{Re}(\lambda) > 0$ for both roots $\Rightarrow$ **unstable spiral**. The equilibrium is no longer stable.

(g) Phase portrait for $\mu = 1$

This system, $\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$, is the Van der Pol oscillator. Flipping the sign of μ reverses the sign of the effective damping near the origin: for $|x| < 1$ the term $\mu(1 - x^2)\dot{x}$ now pumps energy into the system instead of removing it, so the origin switches from a sink to a source. But since $t \to -t$ maps this system into the $\mu = -1$ system of (c), the change is more than just the origin flipping sign: *both* the origin and the limit cycle flip stability together. At $\mu = -1$ the cycle is the unstable outer boundary of the origin's basin (part (c)); at $\mu = 1$ that same cycle becomes stable, and it is now the origin that is unstable, with trajectories spiraling into the cycle from both inside and outside.

This is also worth naming as a Hopf bifurcation in μ, but a degenerate one, not the textbook supercritical case. At $\mu = 0$ the system is simply $\ddot{x} + x = 0$: a center, with a whole plane of neutrally stable periodic orbits at every amplitude, not an isolated equilibrium. In a generic supercritical Hopf bifurcation, the limit cycle is born at the bifurcation point with zero amplitude and grows continuously as $\sqrt{\mu}$; here instead the cycle amplitude stays ≈ 2 as $\mu \to 0^+$ (and $\mu \to 0^-$), rather than shrinking to zero at $\mu = 0$.

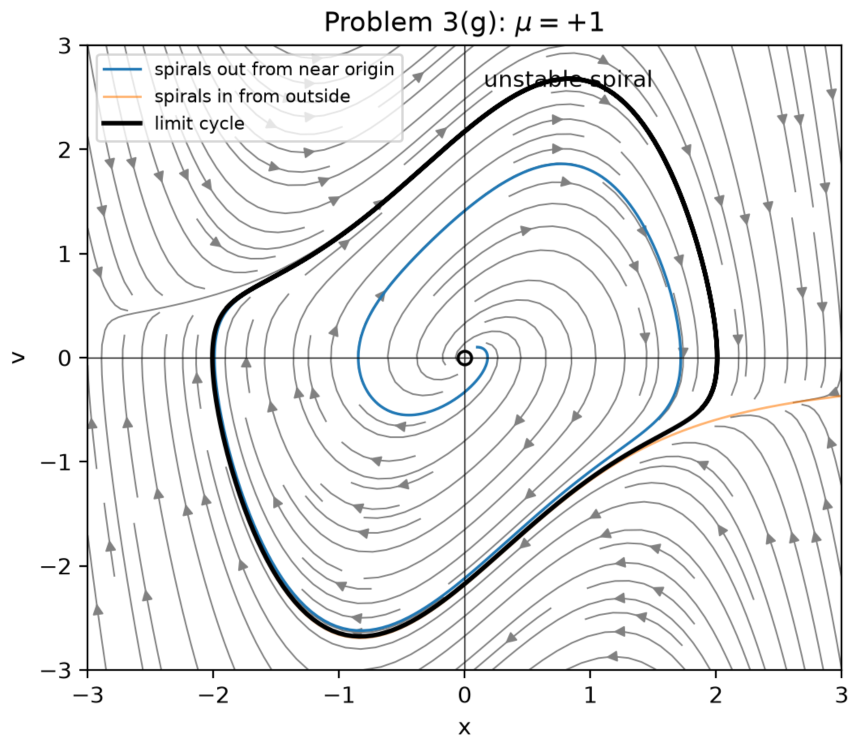


Figure 6: Phase portrait for Problem 3(g): $\mu = 1$, the Van der Pol oscillator. The origin is now unstable, and the limit cycle (amplitude ≈ 2, the time-reversed image of the unstable cycle in Fig. 5) is stable, attracting trajectories from both inside and outside.

Problem 4

Problem 4

Consider a mass hanging vertically from a nonlinear spring, with a total potential energy given by

$$V(y) = -mgy - \frac{1}{2}k_1 y^2 + \frac{1}{4}k_2 y^4$$

When k_1 is positive, the spring is repelling for small values of y, and when it is negative, it is attracting. y is defined positive downward. This is similar to the pitchfork bifurcation example that we saw in 2D.

The system also has a damping force, $F_d = -b\dot{y}$. This gives the equation of motion,

$$m\ddot{y} = -b\dot{y} - \frac{dV}{dy}$$

(a) Nondimensionalize the system. Show that by choosing your time and length scalings to be $T = \frac{m}{b}$ and \(L=\frac{b}{\sqrt{k_2 m}\), you reduce to the system,

$$\ddot{x} + \dot{x} - ax + x^3 = h,$$

where $x = \frac{y}{L}$. What are a, h?

(b) Plot two phase potraits, one with h=0.01, a=-1, and the other with h=0.01, a=1. What do you notice about those phase potraits?

(c) Convert the system to (x, v) state space. Equilibria occur only when $v^* = 0$. What is the cubic polynomial that needs to be solved to find x^*?

(d) Analytically calculate the Jacobian of the system.

(e) Using a polynomial root solver (like roots in MATLAB or numpy.roots in Python), generate a scatter plot of all real roots of the system over the range $a \in [-1, 1]$ for h=0.01. Use an eigenvalue solver to solve for the eigenvalues of the system to classify stability and color the roots according to their stability. You'll notice that this is a perturbed version of the pitchfork bifurcation. Because the addition of a small asymmetry (such as h=0.01) breaks the pitchfork bifurcation, the bifurcation itself is not stable.

(a) Non-dimensionalization

$$\frac{dV}{dy} = -Mg - k_1 y + k_2 y^3 \qquad \Longrightarrow \qquad M\ddot{y} = -b\dot{y} - \frac{dV}{dy} = -b\dot{y} + Mg + k_1 y - k_2 y^3$$

Let $y = Lx$, $t = T\varsigma$, so $\dfrac{d^2y}{dt^2} = \dfrac{L}{T^2}\ddot{x}$ and $\dfrac{dy}{dt} = \dfrac{L}{T}\dot{x}$:

$$\frac{ML}{T^2}\ddot{x} = -\frac{bL}{T}\dot{x} + Mg + k_1 L x - k_2 L^3 x^3$$

$$\ddot{x} = -\frac{bT}{M}\dot{x} + \frac{gT^2}{L} + \frac{k_1 T^2}{M}x - \frac{k_2 L^2 T^2}{M}x^3$$

Choose $\dfrac{bT}{M} = 1 \;\Rightarrow\; T = \dfrac{M}{b}$, then choose $\dfrac{k_2 L^2 T^2}{M} = 1$:

$$L^2 = \frac{M}{k_2 T^2} = \frac{Mb^2}{k_2 M^2} = \frac{b^2}{k_2 M} \qquad \Longrightarrow \qquad L = \frac{b}{\sqrt{k_2 M}}$$

$$\ddot{x} + \dot{x} - \frac{k_1 M}{b^2}x + x^3 = \frac{g\sqrt{k_2}\,M^{5/2}}{b^3}$$

$$\boxed{a = \frac{k_1 M}{b^2}, \qquad h = \frac{g\sqrt{k_2}\,M^{5/2}}{b^3}}$$

(b) Phase portraits, $h = 0.01$

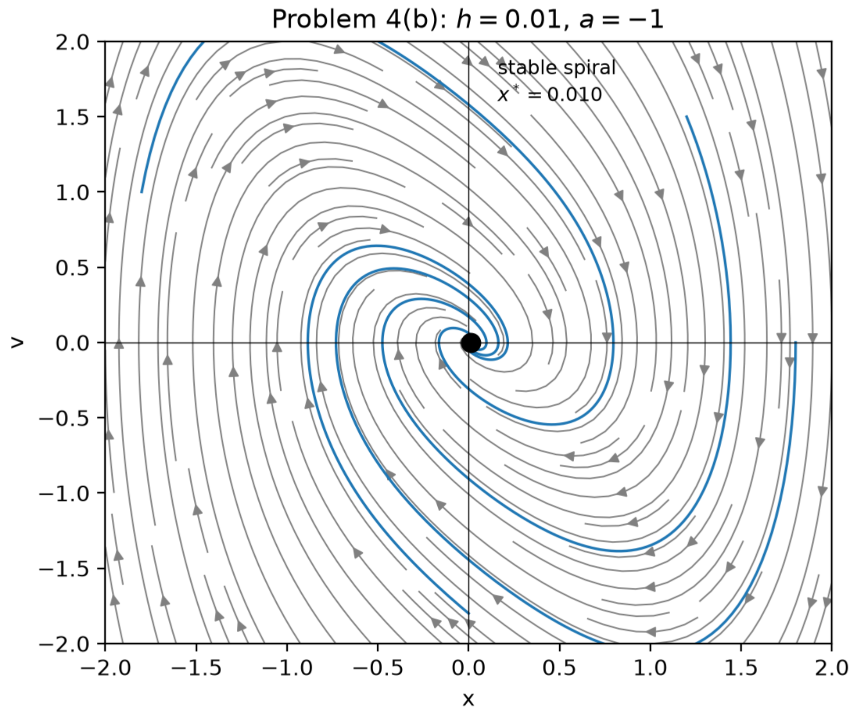


Figure 7: Phase portrait for Problem 4(b): $h = 0.01$, $a = -1$.

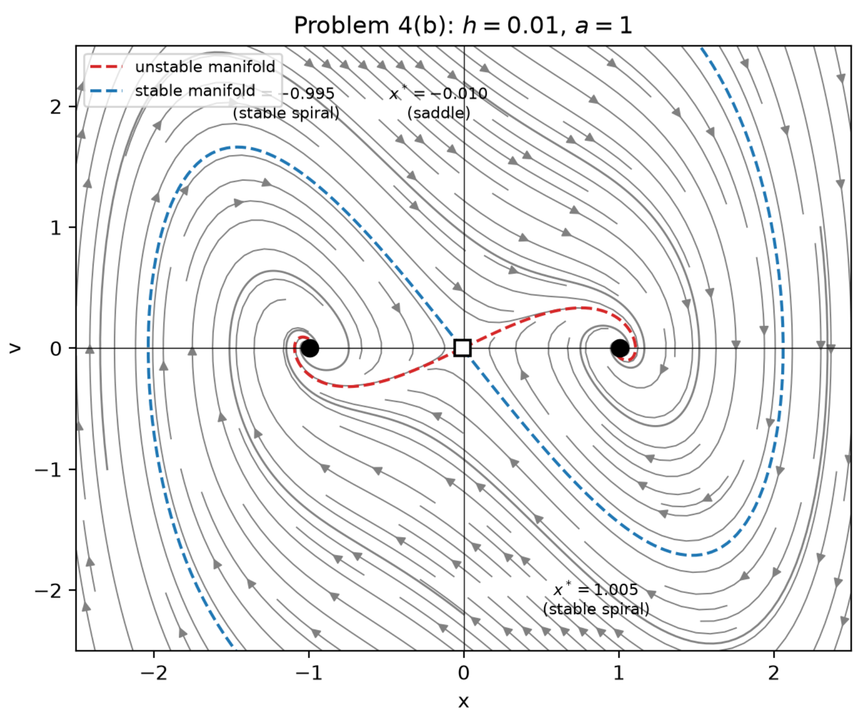


Figure 8: Phase portrait for Problem 4(b): $h = 0.01$, $a = 1$.

For $a = -1$, the equilibrium cubic $x^3 + x - h = 0$ is monotonic ($3x^2 + 1 > 0$ everywhere), so there is exactly one real equilibrium (near $x \approx 0.01$), and the flow settles into it as a single stable spiral. For $a = 1$, the cubic $x^3 - x - h = 0$ picks up three real roots, near $x \approx -0.995$, -0.010, 1.005 for $h = 0.01$: the small asymmetry h breaks the perfect symmetry of the pitchfork, unequally shifting the two flanking equilibria and offsetting the middle root away from exactly zero. The two outer equilibria are stable spirals and the middle one is a saddle (see part (d)).

(c) State-space equilibria

$$\dot{x} = v, \qquad \dot{v} = -v + ax - x^3 + h$$

Fixed points require $v^* = 0$, which leaves:

$$x^{*3} - ax^* - h = 0$$

(d) Jacobian

$$J = \begin{bmatrix} 0 & 1 \\ a - 3x^2 & -1 \end{bmatrix}$$

The trace $\tau = \operatorname{tr} J = -1$ for *every* equilibrium, independent of a, h, or x^*. So no equilibrium of this system can turn unstable through a sign change in the trace: instability can only arise through $\Delta = \det J = 3x^{*2} - a < 0$, i.e. a saddle.

(e) Root-scatter plot over $a \in [-1, 1]$, $h = 0.01$

For each a in the range, the real roots of $x^{*3} - ax^* - h = 0$ (via a polynomial root solver) are the equilibria; each is classified by $\Delta = 3x^{*2} - a$: $\Delta < 0$ is a saddle (unstable), $\Delta > 0$ is stable (a spiral whenever $\tau^2 - 4\Delta = 1 - 4\Delta < 0$, i.e. $\Delta > 1/4$).

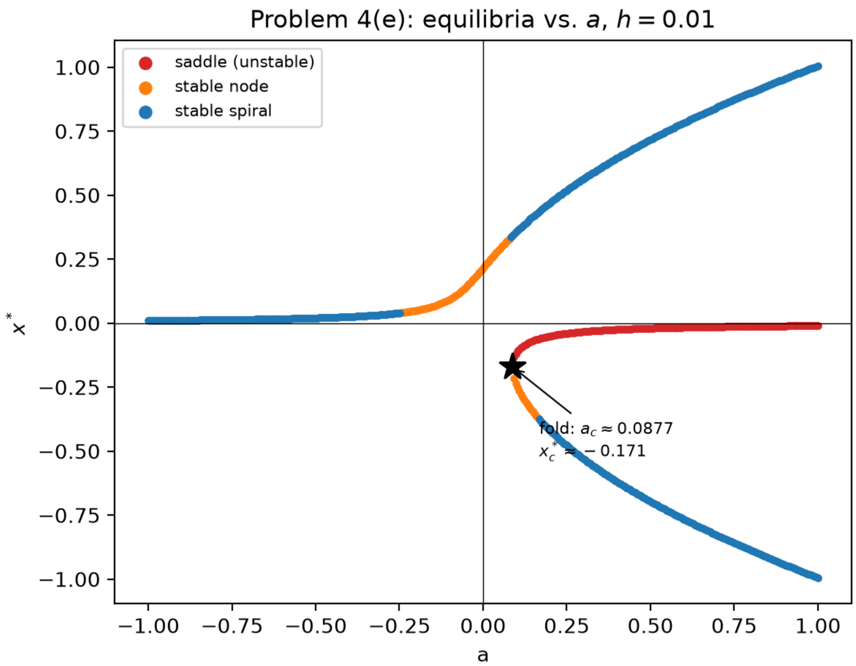


Figure 9: Equilibria x^* vs. a for $h = 0.01$, colored by stability. The fold at $a_c \approx 0.0877$, $x_c^* \approx -0.171$ is marked where the saddle and lower stable branches are born.

Because $h = 0.01 \neq 0$ breaks the exact symmetry of the pitchfork, the branch structure is not the clean $x^* = 0$ and $x^* = \pm\sqrt{a}$ curves of the ideal case. But the surviving branch at $a < 0$ was never going to meet the saddle branch at one point: it continues smoothly through $a = 0$ onto the upper $x^* \to +1$ branch and never bifurcates at all. What actually happens instead is a **saddle-node (fold) bifurcation**: the saddle and the lower stable branch are born together, out of nothing, at the point where the cubic $x^{*3} - ax^* - h = 0$ becomes tangent to the x^*-axis, i.e. exactly where $\Delta = 3x^{*2} - a = 0$ as well. Solving $3x^{*2} = a$ for $x^* = -\sqrt{a/3}$ (the branch consistent with the plot) and substituting into the cubic eliminates x^*:

$$x^{*3} - ax^* - h = 0 \implies x^*\left(\frac{a}{3} - a\right) = h \implies x^*\left(-\frac{2a}{3}\right) = h$$

Substituting $x^* = -\sqrt{a/3}$:

$$-\sqrt{\frac{a}{3}}\left(-\frac{2a}{3}\right) = h \implies \frac{2a}{3}\sqrt{\frac{a}{3}} = h,$$

which for $h = 0.01$ gives

$$a_c \approx 0.0877, \qquad x_c^* = -\sqrt{a_c/3} \approx -0.171,$$

exactly where the two new branches appear in the scatter plot above. Physically, recall $a = k_1 M/b^2$, so $a > 0 \Leftrightarrow k_1 > 0$ is a repelling spring: below a_c that repelling term is too weak to create any equilibria beyond the one branch continuing from $a < 0$; only above a_c is it strong enough to carve out the saddle-node pair.

Appendix: Code Listings

Problem 1 Code

```python
import os
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

FIG_DIR = os.path.join(os.path.dirname(__file__), "..", "latex", "Figures", "Phase Portraits")
os.makedirs(FIG_DIR, exist_ok=True)


def savefig(fig, name):
    fig.savefig(os.path.join(FIG_DIR, name), dpi=200, bbox_inches="tight")


M, k = 1.0, 1.0
omega = np.sqrt(k / M)

X, V = np.meshgrid(np.linspace(-2, 2, 400), np.linspace(-2, 2, 400))
dX = V
dV = -(k / M) * X

fig, ax = plt.subplots(figsize=(6, 5))
ax.streamplot(X, V, dX, dV, density=1.4, color="gray", linewidth=0.7)

theta = np.linspace(0, 2 * np.pi, 200)
for r in (0.5, 1.0, 1.5, 2.0):
    ax.plot(r * np.cos(theta), omega * r * np.sin(theta), color="C0", lw=1.3)

ax.plot(0, 0, "ko", ms=6)
ax.annotate("center", (0, 0), xytext=(0.15, 1.85), fontsize=10)
ax.axhline(0, lw=0.5, c="k")
ax.axvline(0, lw=0.5, c="k")
ax.set_xlabel("x")
ax.set_ylabel("v")
ax.set_title("Problem 1(a): frictionless sliding")
ax.set_xlim(-2, 2)
ax.set_ylim(-2, 2)
savefig(fig, "Problem1a.png")
plt.close(fig)


b_cr = 2 * np.sqrt(k * M)


def damped_rhs(t, state, b):
    x, v = state
    return [v, -(k / M) * x - (b / M) * v]


fig, (ax_spiral, ax_crit, ax_node) = plt.subplots(1, 3, figsize=(16, 5))

b_spiral = 0.5 * b_cr
X, V = np.meshgrid(np.linspace(-2, 2, 400), np.linspace(-2, 2, 400))
dX = V
dV = -(k / M) * X - (b_spiral / M) * V
```

```python

ax_spiral.streamplot(X, V, dX, dV, density=1.4, color="gray", linewidth=0.7)

for x0 in ([1.8, 0], [-1.5, 1.0], [0, -1.8]):
    sol = solve_ivp(damped_rhs, [0, 30], x0, args=(b_spiral,), max_step=0.02)
    ax_spiral.plot(sol.y[0], sol.y[1], color="C0", lw=1.3)

ax_spiral.plot(0, 0, "ko", ms=6)
ax_spiral.annotate("stable spiral", (0, 0), xytext=(0.15, 1.85), fontsize=10)
ax_spiral.axhline(0, lw=0.5, c="k")
ax_spiral.axvline(0, lw=0.5, c="k")
ax_spiral.set_xlabel("x")
ax_spiral.set_ylabel("v")
ax_spiral.set_title(rf"$b={b_spiral:.2g} < b_{{cr}}={b_cr:.2g}$ (spiral)")
ax_spiral.set_xlim(-2, 2)
ax_spiral.set_ylim(-2, 2)

lam_crit = -np.sqrt(k / M)

X, V = np.meshgrid(np.linspace(-2, 2, 400), np.linspace(-2, 2, 400))
dX = V
dV = -(k / M) * X - (b_cr / M) * V

ax_crit.streamplot(X, V, dX, dV, density=1.4, color="gray", linewidth=0.7)

xs = np.linspace(-2, 2, 50)
ax_crit.plot(xs, lam_crit * xs, "--", color="C1", lw=1.5,
             label=f"coalesced eigendirection ($\\lambda={lam_crit:.2f}$)")

for x0 in ([1.8, -0.5], [-1.8, 0.5], [1.0, 1.8], [-1.0, -1.8]):
    sol = solve_ivp(damped_rhs, [0, 30], x0, args=(b_cr,), max_step=0.02)
    ax_crit.plot(sol.y[0], sol.y[1], color="C0", lw=1.2)

ax_crit.plot(0, 0, "ko", ms=6)
ax_crit.annotate("degenerate node", (0, 0), xytext=(0.15, 1.85), fontsize=10)
ax_crit.axhline(0, lw=0.5, c="k")
ax_crit.axvline(0, lw=0.5, c="k")
ax_crit.set_xlabel("x")
ax_crit.set_ylabel("v")
ax_crit.set_title(rf"$b=b_{{cr}}={b_cr:.2g}$ (critically damped)")
ax_crit.set_xlim(-2, 2)
ax_crit.set_ylim(-2, 2)
ax_crit.legend(loc="lower right", fontsize=8)

b_node = 1.5 * b_cr
lam = np.roots([1, b_node / M, k / M])
lam_slow, lam_fast = sorted(lam, key=abs)

X, V = np.meshgrid(np.linspace(-2, 2, 400), np.linspace(-2, 2, 400))
dX = V
dV = -(k / M) * X - (b_node / M) * V

ax_node.streamplot(X, V, dX, dV, density=1.4, color="gray", linewidth=0.7)

xs = np.linspace(-2, 2, 50)
ax_node.plot(xs, lam_slow * xs, "--", color="C1", lw=1.5, label=f"slow eigendirection ($\\lambda={
    lam_slow:.2f}$)")
ax_node.plot(xs, lam_fast * xs, "--", color="C3", lw=1.5, label=f"fast eigendirection ($\\lambda={
    lam_fast:.2f}$)")
```

```
for x0 in ([1.8, -0.5], [-1.8, 0.5], [1.0, 1.8], [-1.0, -1.8]):
    sol = solve_ivp(damped_rhs, [0, 30], x0, args=(b_node,), max_step=0.02)
    ax_node.plot(sol.y[0], sol.y[1], color="C0", lw=1.2)

ax_node.plot(0, 0, "ko", ms=6)
ax_node.annotate("stable node", (0, 0), xytext=(0.15, 1.85), fontsize=10)
ax_node.axhline(0, lw=0.5, c="k")
ax_node.axvline(0, lw=0.5, c="k")
ax_node.set_xlabel("x")
ax_node.set_ylabel("v")
ax_node.set_title(rf"$b={b_node:.2g} > b_{{cr}}={b_cr:.2g}$ (node)")
ax_node.set_xlim(-2, 2)
ax_node.set_ylim(-2, 2)
ax_node.legend(loc="lower right", fontsize=8)

fig.suptitle("Problem 1(b): linear damper")
fig.tight_layout()
savefig(fig, "Problem1b.png")
plt.close(fig)


muN = 1.0

def coulomb_rhs(t, state):
    x, v = state
    if abs(v) < 1e-3 and abs(k * x) <= muN:
        return [0.0, 0.0]
    fric = muN if v > 0 else (-muN if v < 0 else 0.0)
    return [v, -(k / M) * x - fric / M]


X, V = np.meshgrid(np.linspace(-6, 6, 500), np.linspace(-5, 5, 500))
dX = V
dV = -(k / M) * X - (muN / M) * np.sign(V)

fig, ax = plt.subplots(figsize=(7, 5.5))
ax.streamplot(X, V, dX, dV, density=1.4, color="gray", linewidth=0.7)

ax.axvspan(-muN / k, muN / k, color="0.85", zorder=0)
ax.annotate("dead band", (0, 4.4), ha="center", fontsize=9)

for x0 in ([5.5, 0.0], [-5.5, 0.0]):
    sol = solve_ivp(coulomb_rhs, [0, 60], x0, max_step=0.005, dense_output=False)
    ax.plot(sol.y[0], sol.y[1], color="C0", lw=1.3)

ax.plot([-muN / k, muN / k], [0, 0], "ko", ms=5)
ax.annotate(r"centers at $\pm\mu N/k$", (muN / k, 0), xytext=(2.3, -3.0), fontsize=9,
            arrowprops=dict(arrowstyle="->", lw=0.8))
ax.axhline(0, lw=0.5, c="k")
ax.axvline(0, lw=0.5, c="k")
ax.set_xlabel("x")
ax.set_ylabel("v")
ax.set_title("Problem 1(c): Coulomb friction")
ax.set_xlim(-6, 6)
ax.set_ylim(-5, 5)
savefig(fig, "Problem1c.png")
plt.close(fig)
```

```python
print(f"Saved Problem1a/1b/1c.png to: {os.path.abspath(FIG_DIR)}")
```

Listing 1: Problem 1 Phase Portraits

Problem 2 Code

```python
import os
import numpy as np
import matplotlib.pyplot as plt

FIG_DIR = os.path.join(os.path.dirname(__file__), "..", "latex", "Figures", "Phase Portraits")
os.makedirs(FIG_DIR, exist_ok=True)


def savefig(fig, name):
    fig.savefig(os.path.join(FIG_DIR, name), dpi=200, bbox_inches="tight")


U, V = np.meshgrid(np.linspace(-1.2, 3.2, 400), np.linspace(-2.5, 2.5, 400))
dU = V
dV = U - 1.0

fig, ax = plt.subplots(figsize=(7, 5.5))
ax.streamplot(U, V, dU, dV, density=1.4, color="gray", linewidth=0.7)

ax.axvspan(-1.2, 0, color="0.85", zorder=0)
ax.annotate("nonphysical\n(u < 0)", (-0.6, -1.3), ha="center", fontsize=8, style="italic")
ax.axvline(0, lw=1.0, c="k")
ax.annotate("bottom of tube, r = 0", (0.05, -2.3), fontsize=8)

us = np.linspace(-1.2, 3.2, 50)
ax.plot(us, (us - 1), "--", color="C3", lw=1.5, label=r"unstable manifold $v=u-1$ ($\lambda=+1$)")
ax.plot(us, -(us - 1), "--", color="C0", lw=1.5, label=r"stable manifold $v=-(u-1)$ ($\lambda=-1$)")

ax.plot(1, 0, "ks", ms=7)
ax.annotate("saddle (1, 0)", (1, 0), xytext=(1.15, 0.25), fontsize=9)

u0, v0_crit = 2.0, -1.0
ax.plot(u0, v0_crit, "o", color="black", ms=6)
ax.annotate(r"$(u_0, v_0) = (2, -1)$" + "\ncritical inward speed", (u0, v0_crit),
            xytext=(1.9, -2.1), fontsize=8, arrowprops=dict(arrowstyle="->", lw=0.8))

t = np.linspace(0, 4, 400)
for v0, color, label in (
    (-1.3, "C1", "faster than critical: reaches r = 0"),
    (-1.0, "black", "exactly critical: asymptotes to saddle"),
    (-0.7, "C2", "slower than critical: turns back"),
):
    A = (u0 - 1 + v0) / 2
    B = (u0 - 1 - v0) / 2
    u_t = 1 + A * np.exp(t) + B * np.exp(-t)
    v_t = A * np.exp(t) - B * np.exp(-t)
    mask = (u_t >= -1.2) & (u_t <= 3.2) & (v_t >= -2.5) & (v_t <= 2.5)
    ax.plot(u_t[mask], v_t[mask], color=color, lw=1.6, label=label)

ax.set_xlabel("u")
```

```
ax.set_ylabel("v")
ax.set_title("Problem 2(c): non-dimensionalized rotating-bead system (saddle)")
ax.set_xlim(-1.2, 3.2)
ax.set_ylim(-2.5, 2.5)
ax.legend(loc="upper left", fontsize=7.5)
savefig(fig, "Problem2c.png")
plt.close(fig)

print(f"Saved Problem2c.png to: {os.path.abspath(FIG_DIR)}")
```

Listing 2: Problem 2 Phase Portrait

Problem 3 Code

```
import os
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

FIG_DIR = os.path.join(os.path.dirname(__file__), "..", "latex", "Figures", "Phase Portraits")
os.makedirs(FIG_DIR, exist_ok=True)


def savefig(fig, name):
    fig.savefig(os.path.join(FIG_DIR, name), dpi=200, bbox_inches="tight")


def rhs(t, state, mu):
    x, v = state
    return [v, -x + mu * (1 - x**2) * v]


mu = -1.0
X, V = np.meshgrid(np.linspace(-3, 3, 400), np.linspace(-3, 3, 400))
dX = V
dV = -X + mu * (1 - X**2) * V

fig, ax = plt.subplots(figsize=(6, 5))
ax.streamplot(X, V, dX, dV, density=1.4, color="gray", linewidth=0.7)

# Seeds strictly inside the (unstable) limit cycle so they genuinely spiral
# into the origin instead of silently escaping to infinity.
for x0 in ([1.5, 0], [-1.2, 0.8], [0, -1.5]):
    sol = solve_ivp(rhs, [0, 30], x0, args=(mu,), max_step=0.02)
    ax.plot(sol.y[0], sol.y[1], color="C0", lw=1.3)

# The unstable limit cycle itself: integrate backwards in time from a point
# near the origin. Forward in time the origin is attracting, so backward in
# time the trajectory is repelled outward and converges onto the cycle.
lc = solve_ivp(rhs, [0, -60], [0.1, 0.0], args=(mu,), max_step=0.02)
tail = lc.t < -50
ax.plot(lc.y[0, tail], lc.y[1, tail], "k", lw=2.0, label="unstable limit cycle")

ax.plot(0, 0, "ko", ms=6)
ax.annotate("locally stable spiral", (0, 0), xytext=(0.15, 2.15), fontsize=10)
ax.annotate("unstable limit cycle\n(amplitude " + r"$\approx2.01$)", (2.01, 0),
            xytext=(0.9, -2.35), fontsize=9, arrowprops=dict(arrowstyle="->", lw=0.8))
```

```
ax.axhline(0, lw=0.5, c="k")
ax.axvline(0, lw=0.5, c="k")
ax.set_xlabel("x")
ax.set_ylabel("v")
ax.set_title(r"Problem 3(c): $\mu=-1$")
ax.set_xlim(-3, 3)
ax.set_ylim(-3, 3)
ax.legend(loc="upper left", fontsize=8)
savefig(fig, "Problem3c.png")
plt.close(fig)


mu = 1.0
X, V = np.meshgrid(np.linspace(-3, 3, 400), np.linspace(-3, 3, 400))
dX = V
dV = -X + mu * (1 - X**2) * V

fig, ax = plt.subplots(figsize=(6, 5))
ax.streamplot(X, V, dX, dV, density=1.4, color="gray", linewidth=0.7)

sol_in = solve_ivp(rhs, [0, 30], [0.1, 0.1], args=(mu,), max_step=0.02)
ax.plot(sol_in.y[0], sol_in.y[1], color="C0", lw=1.2, label="spirals out from near origin")

sol_out = solve_ivp(rhs, [0, 30], [3.0, 3.0], args=(mu,), max_step=0.02)
ax.plot(sol_out.y[0], sol_out.y[1], color="C1", lw=1.0, alpha=0.6, label="spirals in from outside")

tail = sol_out.t > 20
ax.plot(sol_out.y[0, tail], sol_out.y[1, tail], color="k", lw=2.0, label="limit cycle")

ax.plot(0, 0, "wo", ms=6, mec="k", mew=1.3)
ax.annotate("unstable spiral", (0, 0), xytext=(0.15, 2.6), fontsize=10)
ax.axhline(0, lw=0.5, c="k")
ax.axvline(0, lw=0.5, c="k")
ax.set_xlabel("x")
ax.set_ylabel("v")
ax.set_title(r"Problem 3(g): $\mu=+1$")
ax.set_xlim(-3, 3)
ax.set_ylim(-3, 3)
ax.legend(loc="upper left", fontsize=8)
savefig(fig, "Problem3g.png")
plt.close(fig)

print(f"Saved Problem3c/3g.png to: {os.path.abspath(FIG_DIR)}")
```

Listing 3: Problem 3 Phase Portraits

Problem 4 Code

```
import os
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

FIG_DIR = os.path.join(os.path.dirname(__file__), "..", "latex", "Figures", "Phase Portraits")
os.makedirs(FIG_DIR, exist_ok=True)

h = 0.01
```

```python

STYLE = {
    "saddle": dict(marker="s", mfc="white", mec="k", label="saddle (unstable)"),
    "stable node": dict(marker="o", mfc="0.5", mec="k", label="stable node"),
    "stable spiral": dict(marker="o", mfc="k", mec="k", label="stable spiral"),
}


def savefig(fig, name):
    fig.savefig(os.path.join(FIG_DIR, name), dpi=200, bbox_inches="tight")


def rhs(t, state, a):
    x, v = state
    return [v, -v + a * x - x**3 + h]


def equilibria(a, tol=1e-8):
    roots = np.roots([1.0, 0.0, -a, -h])
    real = np.sort(roots[np.abs(roots.imag) < tol].real)
    return real


def classify(xstar, a):
    Delta = 3 * xstar**2 - a
    if Delta < 0:
        return "saddle"
    elif Delta <= 0.25:
        # Delta == 0.25 exactly is the degenerate-node boundary, a
        # measure-zero set of (a, x*) that never actually occurs on the
        # sampled grid; bucketing it with "stable node" is harmless.
        return "stable node"
    else:
        return "stable spiral"


def mark_equilibrium(ax, xstar, kind):
    st = STYLE[kind]
    ax.plot(xstar, 0, st["marker"], mfc=st["mfc"], mec=st["mec"], ms=8, mew=1.3, zorder=5)


a = -1.0
X, V = np.meshgrid(np.linspace(-2, 2, 400), np.linspace(-2, 2, 400))
dX = V
dV = -V + a * X - X**3 + h

fig, ax = plt.subplots(figsize=(6, 5))
ax.streamplot(X, V, dX, dV, density=1.4, color="gray", linewidth=0.7)

roots = equilibria(a)
for xstar in roots:
    kind = classify(xstar, a)
    mark_equilibrium(ax, xstar, kind)
    ax.annotate(f"{kind}\n$x^*={xstar:.3f}$", (xstar, 0), xytext=(xstar + 0.15, 1.6), fontsize=9)

for x0 in ([1.8, 0], [-1.8, 1.0], [0, -1.8], [1.2, 1.5]):
    sol = solve_ivp(rhs, [0, 30], x0, args=(a,), max_step=0.02)
    ax.plot(sol.y[0], sol.y[1], color="C0", lw=1.2)

```

```python
ax.axhline(0, lw=0.5, c="k")
ax.axvline(0, lw=0.5, c="k")
ax.set_xlabel("x")
ax.set_ylabel("v")
ax.set_title(f"Problem 4(b): $h={h}$, $a={a:g}$")
ax.set_xlim(-2, 2)
ax.set_ylim(-2, 2)
savefig(fig, "Problem4b-am1.png")
plt.close(fig)


a = 1.0
X, V = np.meshgrid(np.linspace(-2.5, 2.5, 400), np.linspace(-2.5, 2.5, 400))
dX = V
dV = -V + a * X - X**3 + h

fig, ax = plt.subplots(figsize=(7, 5.5))
ax.streamplot(X, V, dX, dV, density=1.4, color="gray", linewidth=0.7)

roots = equilibria(a)
kinds = [classify(xstar, a) for xstar in roots]
for xstar, kind in zip(roots, kinds):
    mark_equilibrium(ax, xstar, kind)
    ax.annotate(f"$x^*={xstar:.3f}$\n({kind})", (xstar, 0), xytext=(xstar - 0.15, 1.9 if xstar < 0 else -2.2),
                fontsize=8, ha="center")

saddle_idx = kinds.index("saddle")
xs = roots[saddle_idx]
Delta = 3 * xs**2 - a
lam = np.roots([1, 1, Delta])
lam_unstable = lam[lam.real > 0][0].real
lam_stable = lam[lam.real < 0][0].real

eps = 1e-3
for sign in (+1, -1):
    v0 = sign * eps * np.array([1, lam_unstable])
    sol = solve_ivp(rhs, [0, 30], [xs + v0[0], v0[1]], args=(a,), max_step=0.02)
    ax.plot(sol.y[0], sol.y[1], "--", color="C3", lw=1.5,
            label="unstable manifold" if sign == 1 else None)

    v0 = sign * eps * np.array([1, lam_stable])
    sol = solve_ivp(rhs, [0, -30], [xs + v0[0], v0[1]], args=(a,), max_step=0.02)
    ax.plot(sol.y[0], sol.y[1], "--", color="C0", lw=1.5,
            label="stable manifold" if sign == 1 else None)

for x0 in ([2.3, 1.0], [-2.3, -1.0], [0, 2.2], [0, -2.2]):
    sol = solve_ivp(rhs, [0, 30], x0, args=(a,), max_step=0.02)
    ax.plot(sol.y[0], sol.y[1], color="0.3", lw=1.0, alpha=0.7)

ax.axhline(0, lw=0.5, c="k")
ax.axvline(0, lw=0.5, c="k")
ax.set_xlabel("x")
ax.set_ylabel("v")
ax.set_title(f"Problem 4(b): $h={h}$, $a={a:g}$")
ax.set_xlim(-2.5, 2.5)
ax.set_ylim(-2.5, 2.5)
ax.legend(loc="upper left", fontsize=8)
savefig(fig, "Problem4b-ap1.png")
```

```python
plt.close(fig)


a_vals = np.linspace(-1, 1, 400)
points = {"saddle": ([], []), "stable node": ([], []), "stable spiral": ([], [])}
for a_i in a_vals:
    for xstar in equilibria(a_i):
        kind = classify(xstar, a_i)
        points[kind][0].append(a_i)
        points[kind][1].append(xstar)

scatter_colors = {"saddle": "C3", "stable node": "C1", "stable spiral": "C0"}
fig, ax = plt.subplots(figsize=(6.5, 5))
for kind, (a_pts, x_pts) in points.items():
    ax.scatter(a_pts, x_pts, s=8, color=scatter_colors[kind], label=STYLE[kind]["label"])

ax.axhline(0, lw=0.5, c="k")
ax.axvline(0, lw=0.5, c="k")

# Saddle-node (fold) bifurcation point: tangency of x*^3 - a x* - h = 0 with
# Delta = 3x*^2 - a = 0 simultaneously. Eliminating x* = -sqrt(a/3) gives
# h = (2a/3) sqrt(a/3); solve numerically for a_c, then back out x*_c.
from scipy.optimize import brentq
fold_eq = lambda a_: (2 * a_ / 3) * np.sqrt(a_ / 3) - h
a_c = brentq(fold_eq, 1e-6, 1.0)
x_c = -np.sqrt(a_c / 3)
ax.plot(a_c, x_c, "*", color="k", ms=14, zorder=6)
ax.annotate(rf"fold: $a_c\approx{a_c:.4f}$" + "\n" + rf"$x^*_c\approx{x_c:.3f}$",
            (a_c, x_c), xytext=(a_c + 0.08, x_c - 0.35), fontsize=8,
            arrowprops=dict(arrowstyle="->", lw=0.8))

ax.set_xlabel("a")
ax.set_ylabel(r"$x^*$")
ax.set_title(f"Problem 4(e): equilibria vs. $a$, $h={h}$")
ax.legend(loc="upper left", fontsize=8, markerscale=2)
savefig(fig, "Problem4e-roots.png")
plt.close(fig)

print(f"Fold bifurcation: a_c={a_c:.4f}, x*_c={x_c:.4f}")
print(f"Saved Problem4b-am1/4b-ap1/4e-roots.png to: {os.path.abspath(FIG_DIR)}")
```

Listing 4: Problem 4 Phase Portraits and Root Scatter Plot